Chapter 1

Rotorcraft

1.1 Introduction

This chapter moves on from airplane movement to quadcopters! You may here rotor-winged aircraft, drones, or unmanned aerial vehicle, which all refer to similar types of the same structure.

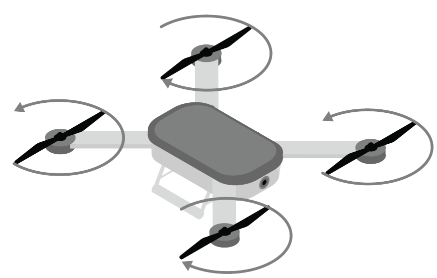


Figure 1.1: The general look of a quadcopter. Includes four rotors, a frame called a fuselage, and generally a head or designated front.

Before we dive in depth, let's introduce a few terms that will be repeated within this chapter.

Quadcopter Often defined as an aircraft with 4 rotors, synonymous with drone.

Rotor The rotor is the part of the aircraft which causes lift by spinning at a high velocity. It consists of a central *hub* with two or more *blades* attached. You may hear the term propeller like from our airplanes chapter, but the more correct term for quadcopters is *rotor*.

UAS / UAV Unmanned Aircraft System or Unmanned Aircraft Vehicle. Often used to describe drones or semi-autonomous quadcopters. This includes the drones themselves, the software needed to avoid obstacles and track positioning, and equipment used by people to communicate with them.

Fuselage The main body of a quadcopter or aircraft; synonymous to frame.

Torque We have not talked about torque yet, but it is an important circular motion concept. Torque is the rotational equivalent of force, and is a measure of how much a force acting on an object causes that object to rotate. For now, understand that it is affects quadcopter movement and is a force that needs to be counteracted in order to maintain stability.

Isometric We also want to be clear about our use of views in our diagrams. An *isometric* perspective is one that visually represents all 3 views of a 3D object in a 2D space (a printed diagram). When we get to calculus, we will talk about *isometry* as transformation or mapping between two spaces that perfectly preserves distances.

1.2 VTOL

The strength of quadcopters lies in their propulsion system. As compared to airplanes, which take off by accelerating down *runways* and gaining enough lift to overcome its weight, quadcopters use spinning rotors to generate this lift. This applies Newton's Third Law — for every action (the downwards lift of the rotors), there is an equal and opposite reaction (the upwards lift of the quadcopter). This allows quadcopters to take off vertically, eliminating the need for a runway. This is a very special type of propulsion known as Vertical Take-Off and Landing (VTOL).

Specifically, the drone's rotors needs to generate enough lift to counteract its own weight and put it into motion in the air. Recall that there are *equal but opposite* forces in play here, so the lift created by the rotors can be found by

$$F_{\text{rotors}} = m_{\text{quadcopter}}\ g + F_{\text{vertical lift}}$$

This is a rudimentary equation that demonstrates the following: the amount of lift created by rotors is the excess force after the rotors are able to overcome the weight of the quadcopter.

Here is an isometric view of vertical flight of a quadcopter

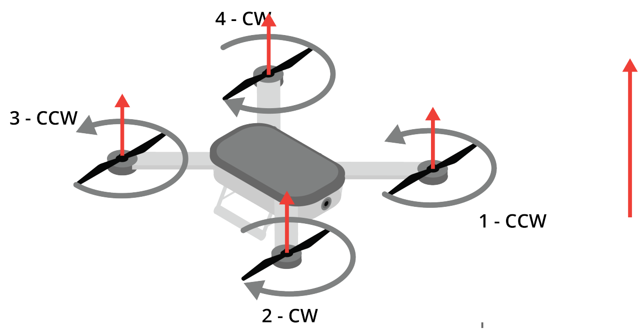


Figure 1.2: A quadcopter taking vertical flight. Notice how the red arrows, representing the strength of the upwards force, are all equivalent.

If we just want our quadcopter to *hover in place*, the forces vertically need to be equal. This means that the weight of the quadcopter will equal the lift generated by the rotors

$$F_{\text{rotors}} = m_{\text{quadcopter}}\ g \tag{1.1}$$

1.3 Rotor Structure

Now, the remaining question is how do rotors work?

Recall our airplanes chapter; the wings are what create lift, and this is heavily dependent on the wings shape. On quadcopters, each rotor is made up of two or more *blades* attached to a central hub, and these blades are specifically designed to push air *opposite* the direction the aircraft needs to go.

[PLACEHOLDER: ISOMETRIC VIEW OF BLADE WITH LABELING]

Each blade has a *root*, which connects to the hub, and a *tip* at its outer end. Along its length, the blade also has a *twist*, where the blade's angle gradually changes from root to tip, similar to the shape of a screw thread.

[PLACEHOLDER: SPLIT FIGURE OF BLADE TWIST COMPARED TO SCREW MECHANISM]

The direction the air is pushed depends on which way the quadcopter goes; reversing the rotation of a rotor reverses the direction of the air pushed, and thus the direction of the quadcopter.

Rotors are typically described by three numbers: their *diameter*, their *pitch*, and their *number of blades*. Typically, you will see something like *10x5 inch*, which refers to the diameter and pitch, and the visual will tell you the number of blades.

The diameter measures the tip to tip distance that a propeller can reach. If there is only 1 blade, this measurement is twice the hub-to-tip distance. Larger diameter rotors move a bigger column of air, but at slower speeds. Thrust scales roughly equal to the square of velocity, while power scales to the cube of velocity. Bigger props are meaningfully more efficient. This is the same reason helicopters use one large slow rotor while quadcopters use several small fast ones.

Two is the typical amount of blades per propeller, and ensure symmetry along the Y-Axis, as well as mass-production capability. Adding blades increases total thrust at a given diameter and RPM, but each additional blade partly flies through the turbulent wake left by the blade ahead of it (this is called *aerodynamic interference*), so you get diminishing thrust returns per blade.

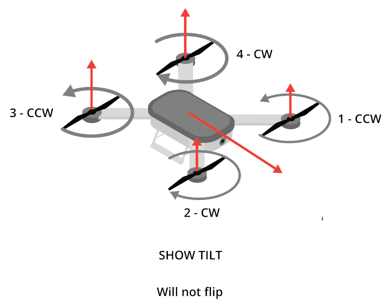
The pitch is the distance the propeller would move after one 360° rotation. Similar to how a screw moves when twisted, this measurement predicts how the propeller would move in vertical space. Lower pitch means less torque needed and higher maximum RPM, but less thrust and vertical movement is produced per rotor. Higher pitch generates more thrust and RPM, but draws more torque from the motor.

1.4 Movement, Spin Direction, and Torque

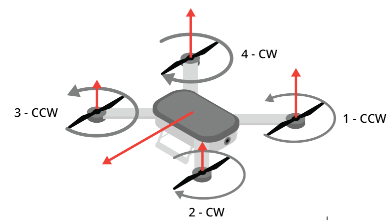
Generally, two rotors spin clockwise (CW) and two spin counter-clockwise (CCW). Rotors positioned diagonally from each other spin in the same direction. Rotors right next to each other spin in opposite directions. If all four rotors spun in the same direction, the body of the drone would violently spin in the opposite direction. By having two CW and two CCW motors, the forces cancel each other out. This balance allows the drone to hover completely still and turn smoothly post-VTOL. There is a guide in your digital resources that goes into blade selection based on your use case, very interesting!

In theory, quadcopters that are symmetrical such as drones could be flown in any direction (this is called *headless flight*). This is easier for beginners as the direction away from the operator is forward, and the direction toward the operator is backwards, regardless of rotation.

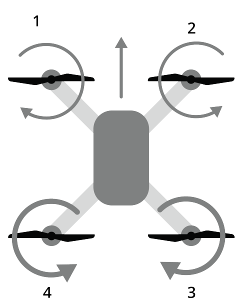
Often a front is decided so that the fuselage body can we can determine roll, pitch, and yaw; this front is called the head. That way, its rotation and flight is based on its relative position in the world, not the operator's location. In the following discussion on forces present on quadcopters, we will assume that there is no drag and our center of gravity is in the center of the fuselage, just as with all other ideal scenarios.



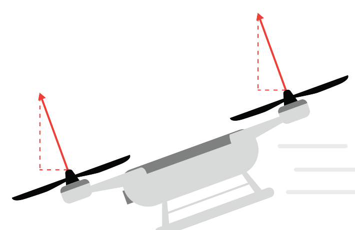
(a) `forwardIso.png`



(b) `lateralIso.png`



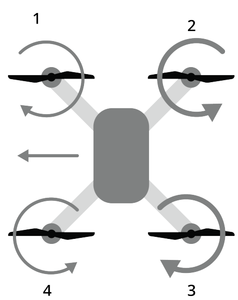
(a) `forwardTop.png`



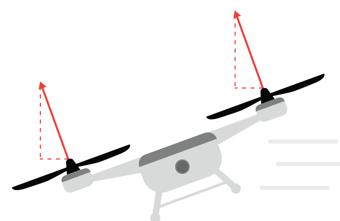
(b) `forwardSide.png`

When flying a quadcopter forward, we apply *pitch* to the quadcopter (figure 1.3a). When doing this, we generate a greater amount of torque from the back two rotors (see 1.4a). This causes the copter to tilt forward with the back rotors higher than the front two rotors (see 1.4b), and direct its motion in the direction of the head. Backwards motion is very similar; the *front* rotors relative to the head spin at a greater rate, and as a result, tilt higher than the back motors, causing backwards motion.

As with airplanes, left and right movement, or *lateral movement*, is controlled by *roll*. Figure 1.3b shows a quadcopter rolling to the left. To produce this roll, the rotors to the *right* of the head spin faster than the rotors to the left, generating extra lift on the right side of the fuselage. This lifts the right side and tips the left side down, tilting the quadcopter's thrust to the left and carrying it in that direction (see 1.5b). Rightward movement works the same way in reverse: the rotors to the *left* spin faster, lifting the left side and rolling the quadcopter to the right.

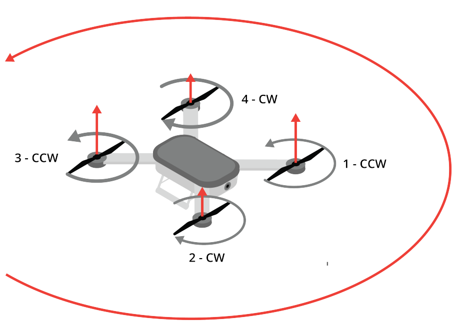


(a) `lateralTop.png`

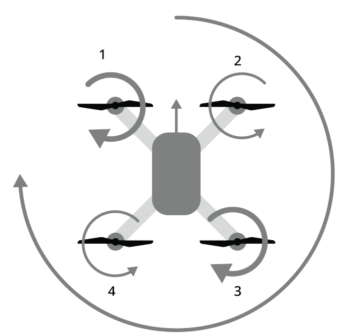


(b) `lateralSide.png`

Rotation of the ahead about the center of gravity is called *yaw*. Instead of two adjacent rotors having increased speeds, we adjust two diagonal rotors to have greater speeds then the other two. Increasing the clockwise rotors will cause a counterclockwise or left rotation. Rightward rotation is similar: increasing the counterclockwise rotors causes a clockwise



(a) `rotationIso.png`



(b) `rotationTop.png`

1.5 Helicopters

Instead of a fixed wing like on an airplane, or four small rotors on a quadcopter, a *helicopter* generates its lift from a single, much larger *main rotor*. The physics is the same as before: the spinning blades push air downward, and Newton's Third Law pushes the aircraft up in response. However, a single big rotor introduces two new problems a quadcopter doesn't have to solve.

First, on a quadcopter, half the rotors spin clockwise and half spin counter-clockwise, so their reaction torques cancel out. A helicopter's main rotor only spins one way, so its reaction torque has nothing to cancel it. Without a fix, the fuselage would spin in the opposite direction of the rotor. Most helicopters solve this with a small *tail rotor* that pushes sideways at a proportional speed to counteract the twist (see Figure 1.7).

Second, a quadcopter changes direction by speeding up or slowing down individual rotors. A helicopter's main rotor spins at a nearly constant speed instead; direction and lift are controlled by changing the *pitch* of the blades as they spin, using controls called *collective* and *cyclic*. We'll cover the tail rotor first, then come back to how collective and cyclic control the main rotor.

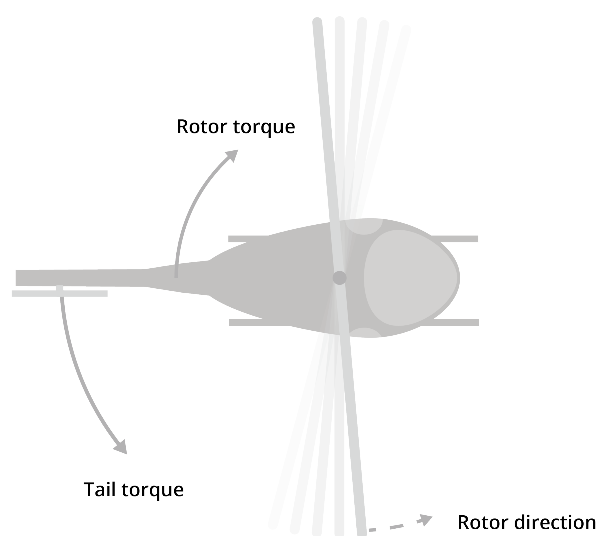
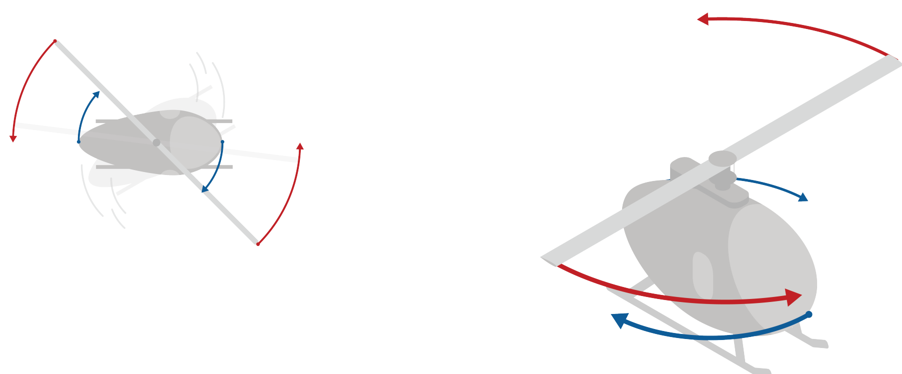


Figure 1.7: A helicopter uses one large rotor that rotates in one direction. A smaller tail rotor counteracts this torque.

1.5.1 Tail Rotor

On a helicopter, the tail rotor spins vertically, pushing air horizontally to counteract the torque of the main rotor. The pilot controls the tail rotor with pedals, which change the pitch of the tail rotor blades. This changes how much air is pushed sideways, and thus how much torque is counteracted.

Let's imagine a helicopter with a main rotor that spins clockwise and *no tail rotor*. Without a tail rotor, the fuselage would have a violent torque applied to it, and would spin counter-clockwise. There is nothing to counteract this torque, so the fuselage would spin faster and faster until the helicopter is destroyed or crashes.



(a) A helicopter without a tail rotor experiences a net torque that spins the fuselage in the opposite direction of the main rotor.

(b) The fuselage continues to rotate faster and faster unless a counter-torque is provided.

Figure 1.8: A helicopter becomes unbalanced without a tail rotor.

Because of this, we need a way to balance this torque. We add a rotor in the back of the helicopter that pushes air in the opposite direction the torque from the main rotor (see Figure 1.9).

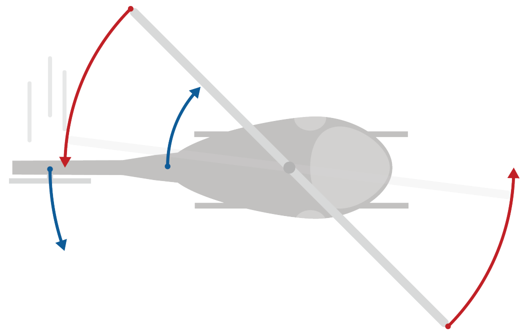


Figure 1.9: A tail rotor provides the counter-torque needed to keep the helicopter balanced and prevent violent movement.

If the pilot wants to yaw left, they increase the pitch of the tail rotor blades, pushing more air to the right and counteracting more of the main rotor's torque. To yaw right, they decrease the pitch of the tail rotor blades, pushing less air to the right and allowing more of the main rotor's torque to spin the fuselage clockwise (see Figure 1.10).

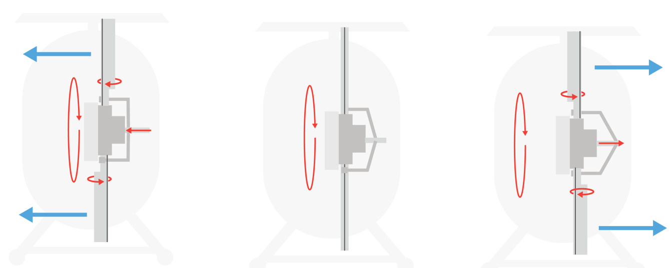


Figure 1.10: The tail rotor changes thrust to control the helicopter's yaw.

1.5.2 Main Rotor Movement

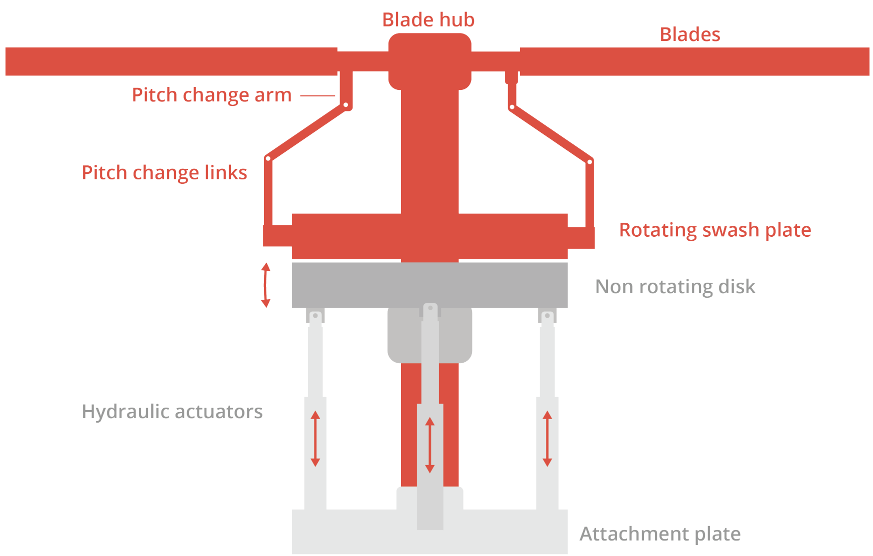


Figure 1.11: `mechanicalControl.png`

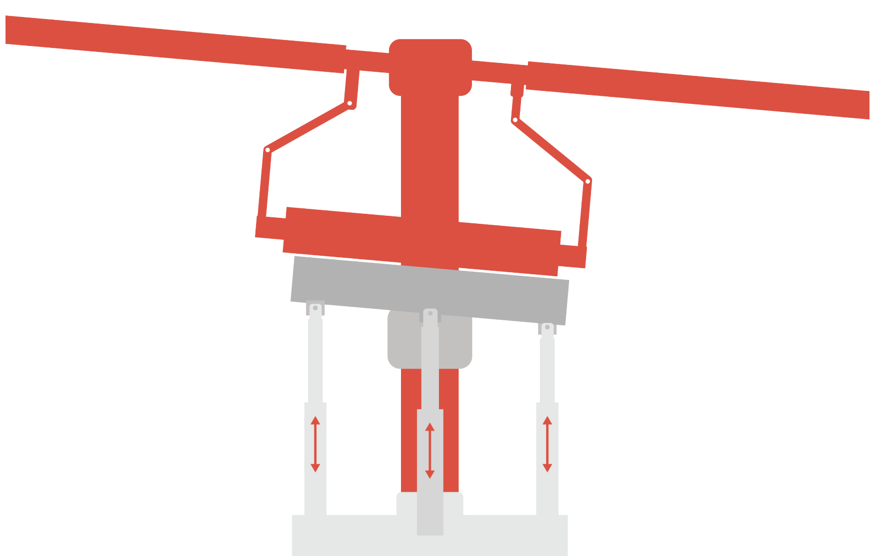


Figure 1.12: `mechanicalControlMovement.png`

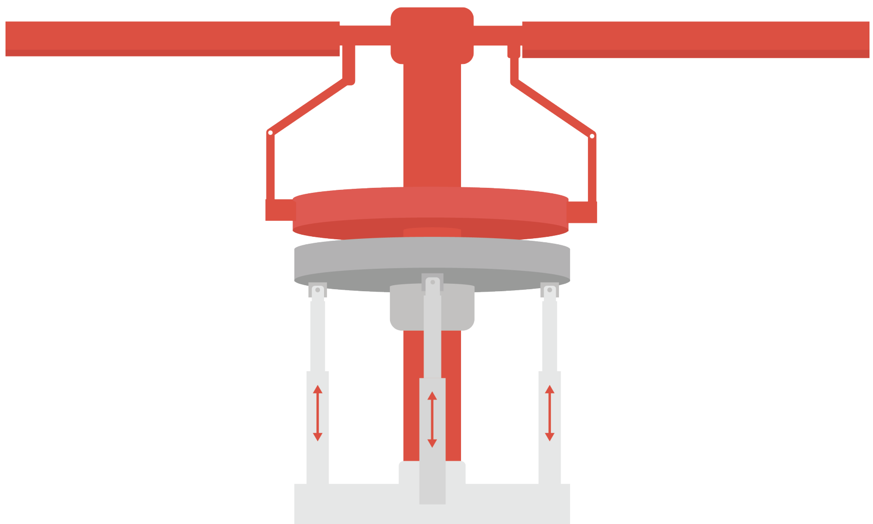


Figure 1.13: `mechanicalControlMovement2.png`

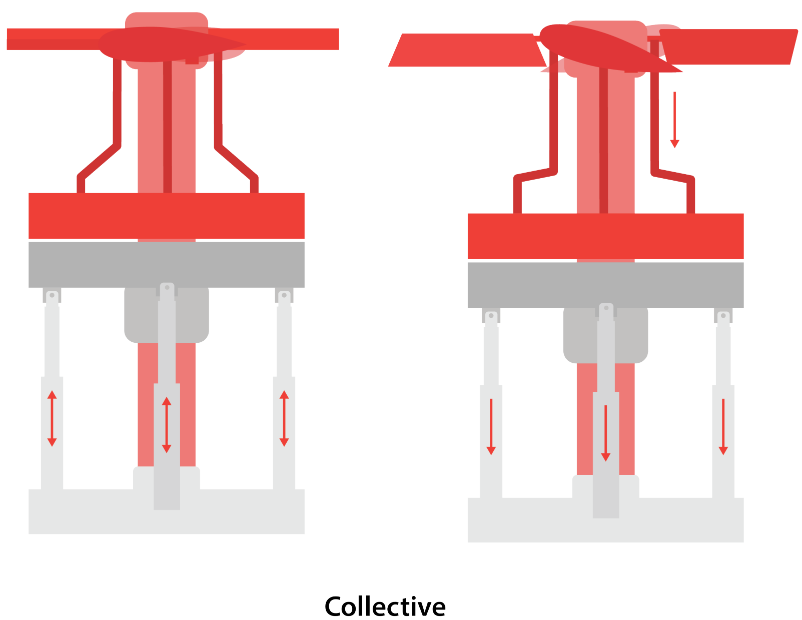


Figure 1.14: `collective.png`

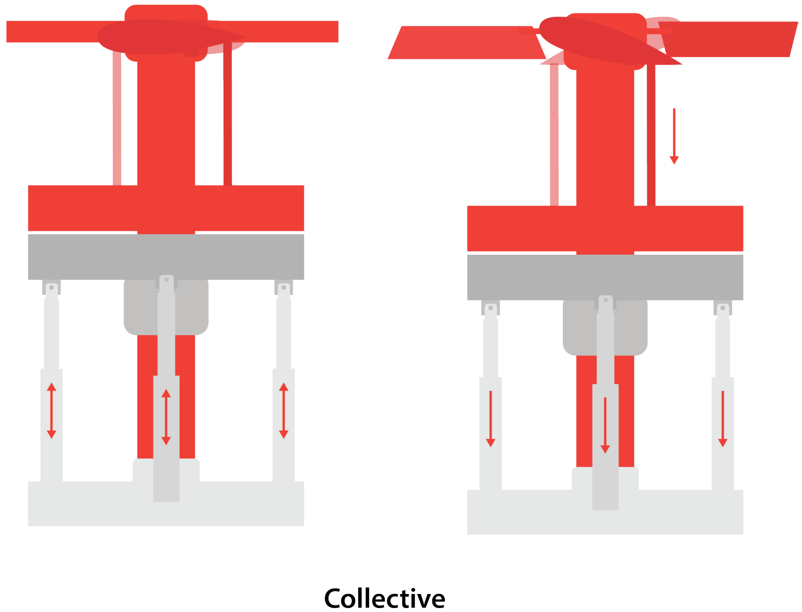


Figure 1.15: `collective4Version.png`

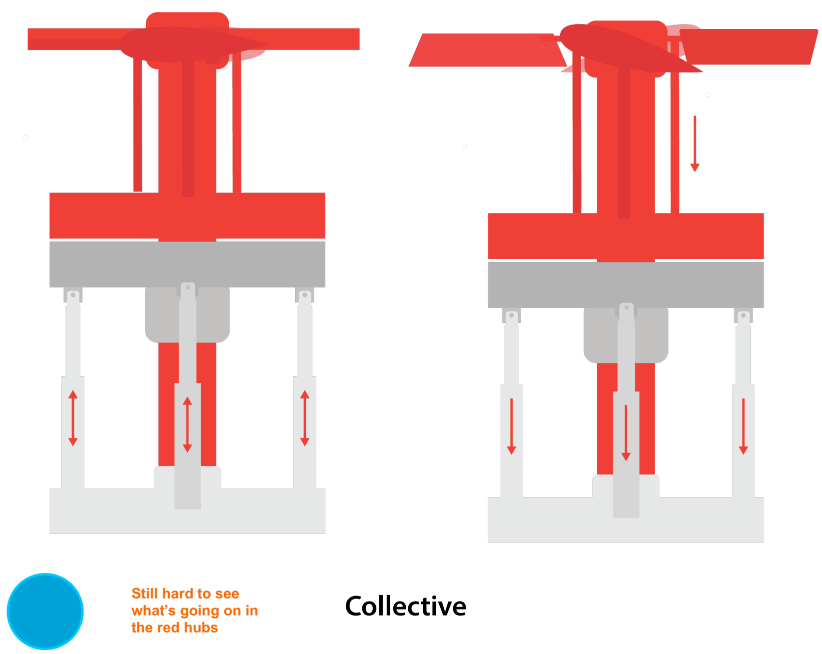


Figure 1.16: `collectiveQuad.png`

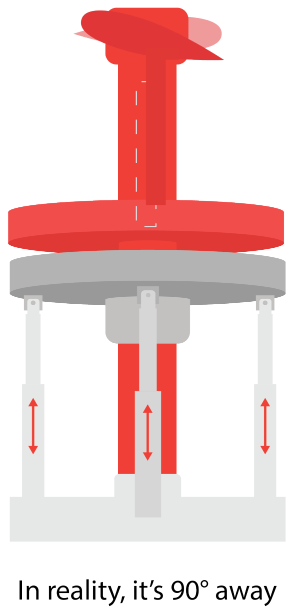


Figure 1.17: `cyclic1.png`

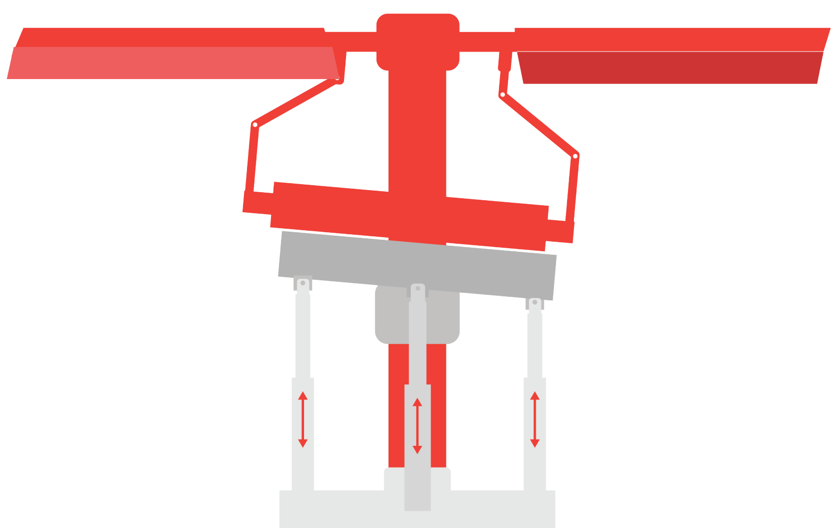


Figure 1.18: `cyclic2.png`

1.5.3 Isometric Copter Views

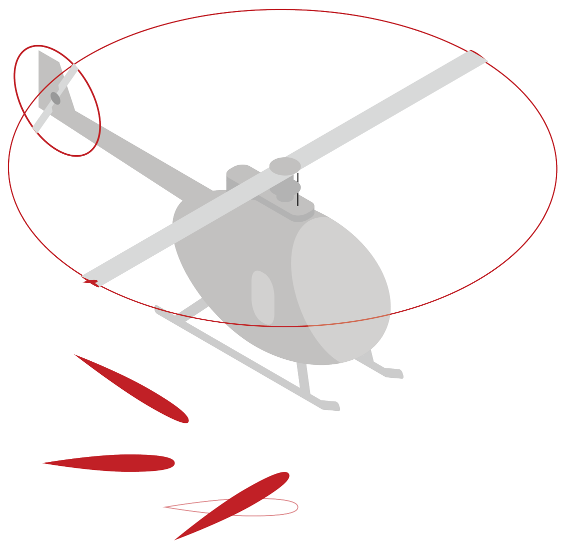


Figure 1.19: `iso-36.png`

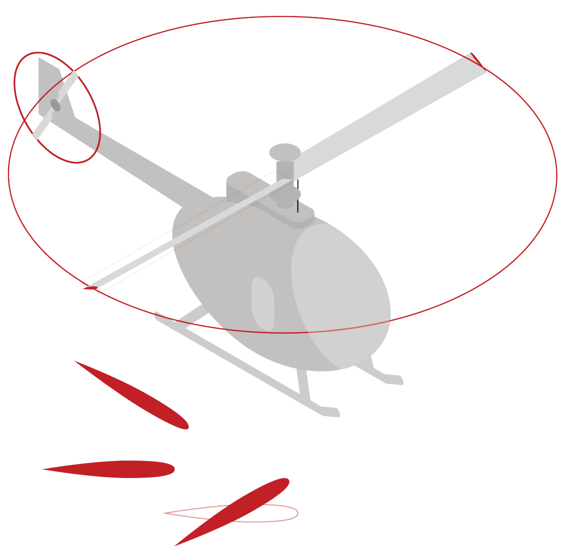


Figure 1.20: `iso-37.png`

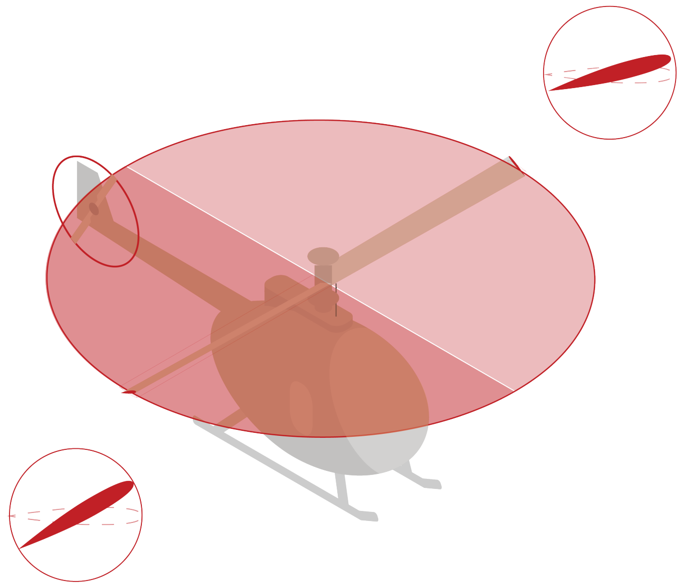


Figure 1.21: iso-38.png

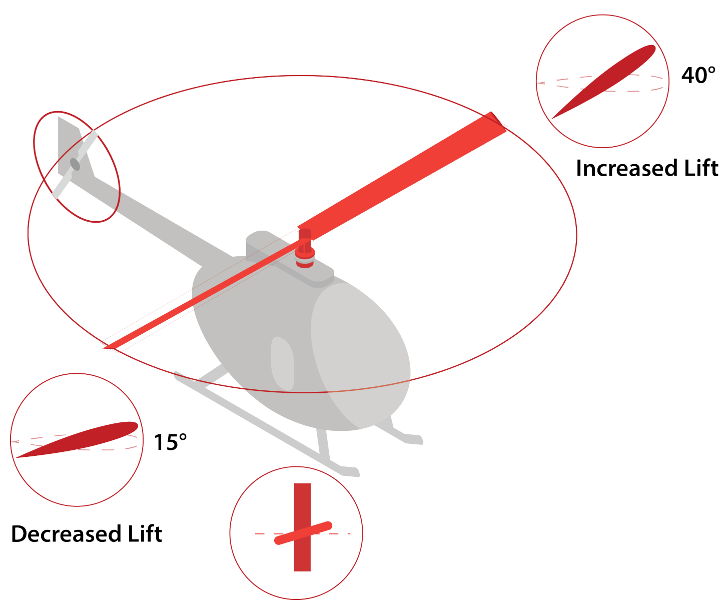


Figure 1.22: iso-48.png

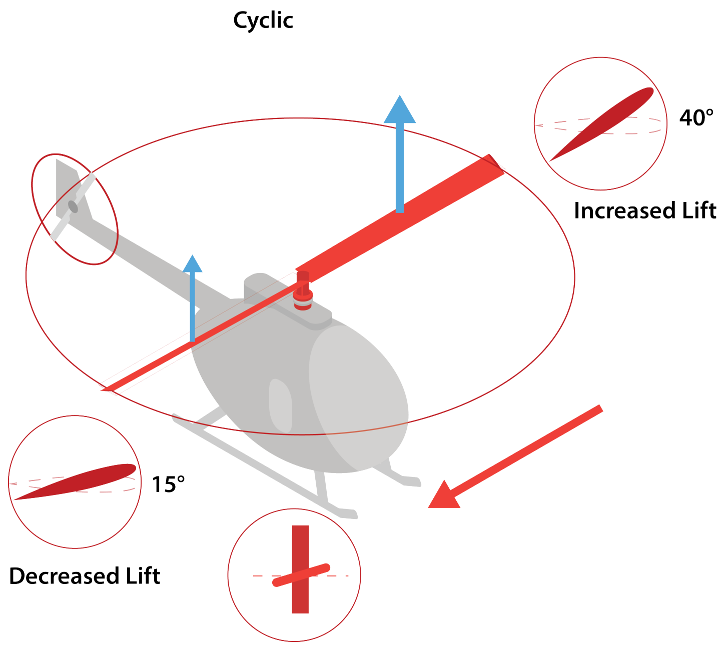


Figure 1.23: iso-48 2.png

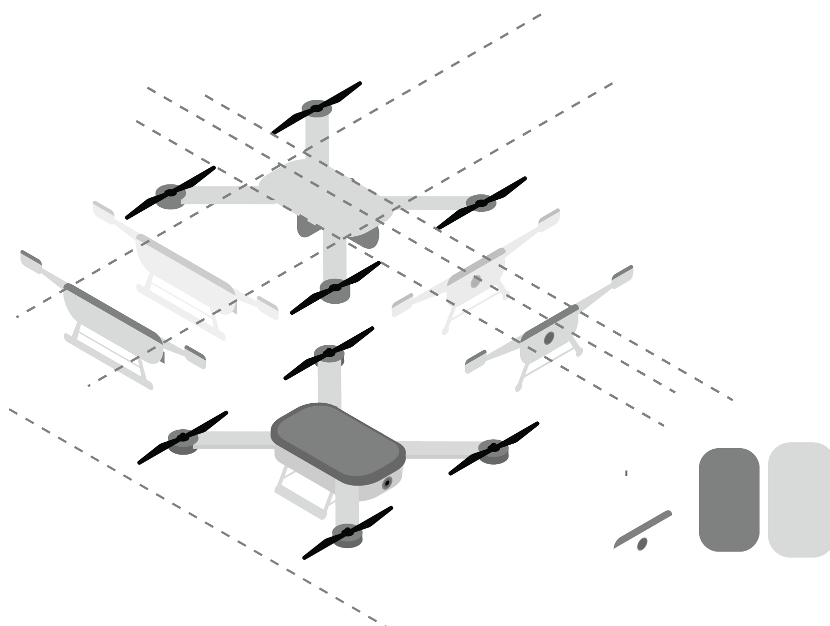


Figure 1.24: iso.png

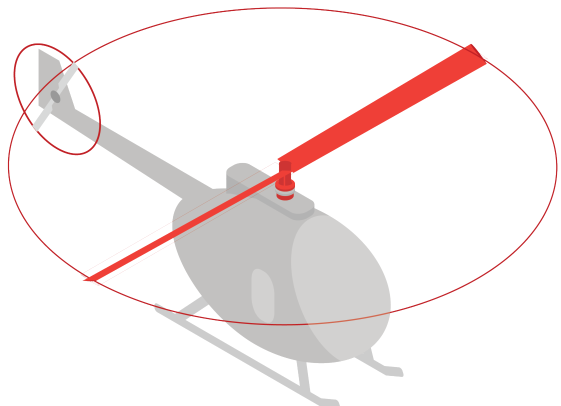


Figure 1.25: `iso46.png`

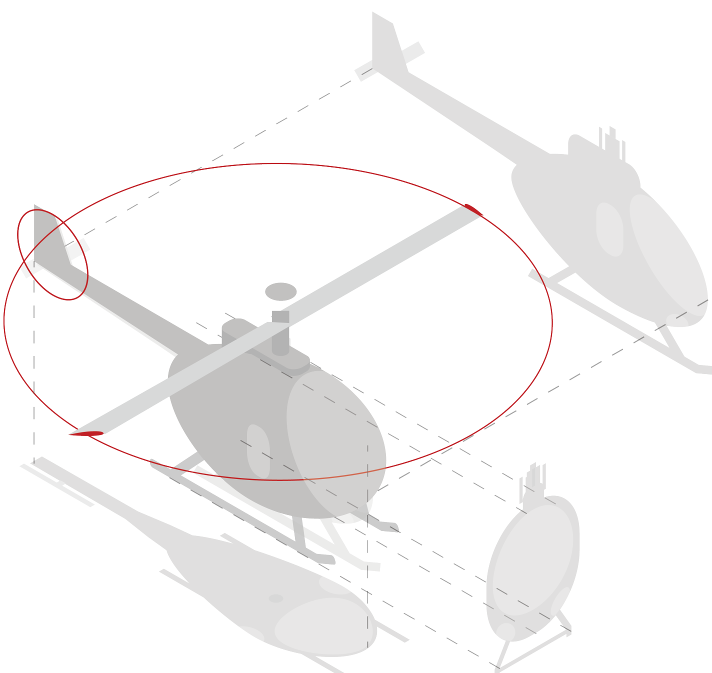


Figure 1.26: `isoprocess.png`

1.5.4 Copter Movement

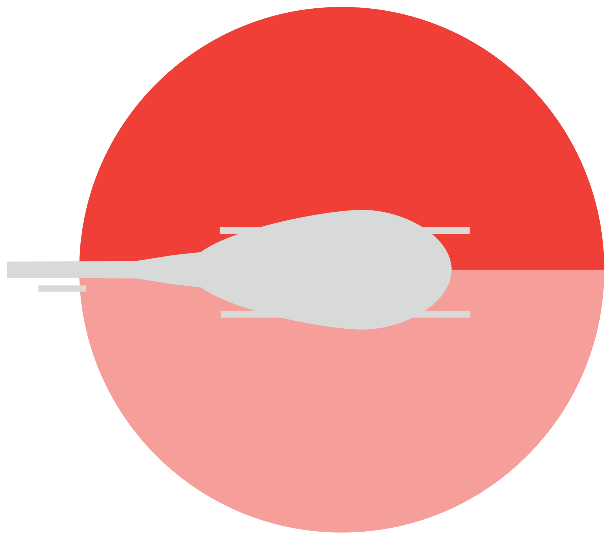


Figure 1.27: `helicopterAbove1.png`

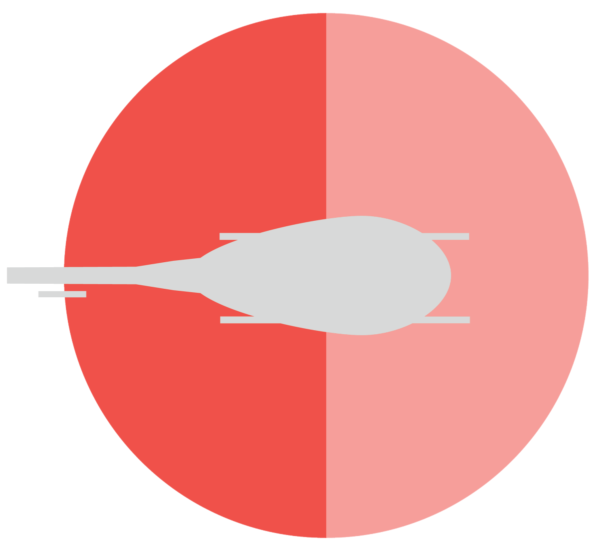


Figure 1.28: `helicopterAbove2.png`

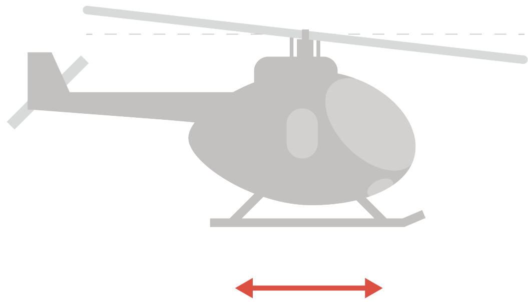


Figure 1.29: `helicopterMovement-10.png`

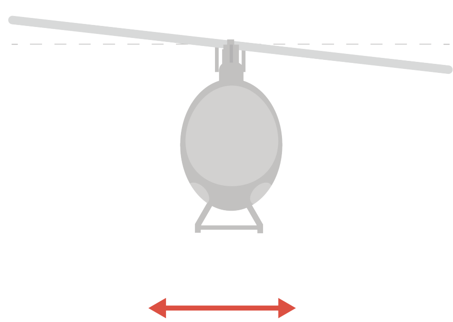


Figure 1.30: `helicopterMovement-11.png`

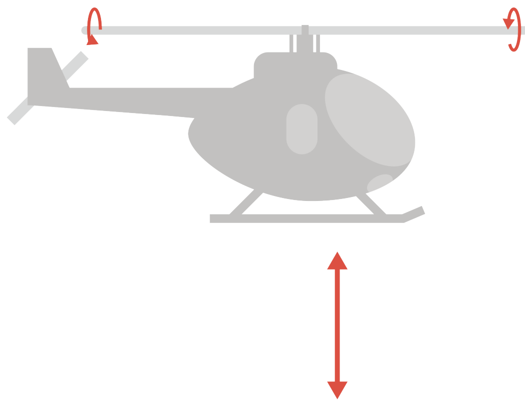


Figure 1.31: `helicopterMovement-12.png`

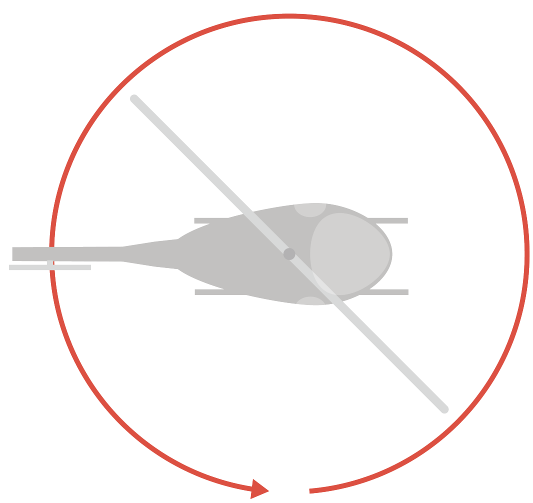


Figure 1.32: `helicopterMovement-13.png`

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Appendix A

Answers to Exercises

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